



English Feature Writing

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The Dark Side of Light: How Bangladesh's Environment is Being Affected by Light Pollution

When the sun went down, humanity used only the stars and moon to guide them as they stumbled through the night. A new discovery, the light bulb, however, revolutionized everything in the late 19th century. The world seemed to become brighter and more linked as individuals were able to brighten their houses and streets at any time of day or night.

The usage of artificial light expanded and took on more forms as technology advanced. Cities were illuminated by skyscrapers, highways were transformed into well-lit thoroughfares, and stadiums for sports were transformed into bright lights at night. It served as a representation of human achievement, invention, and development. But as the years passed, the effects of this excess of light started to become apparent. Many creatures that depended on darkness to survive started to suffer as the night sky once shined with countless stars started to grow darker. This is the negative impact of artificial light, commonly referred to as light pollution, and it has an impact on everyone.

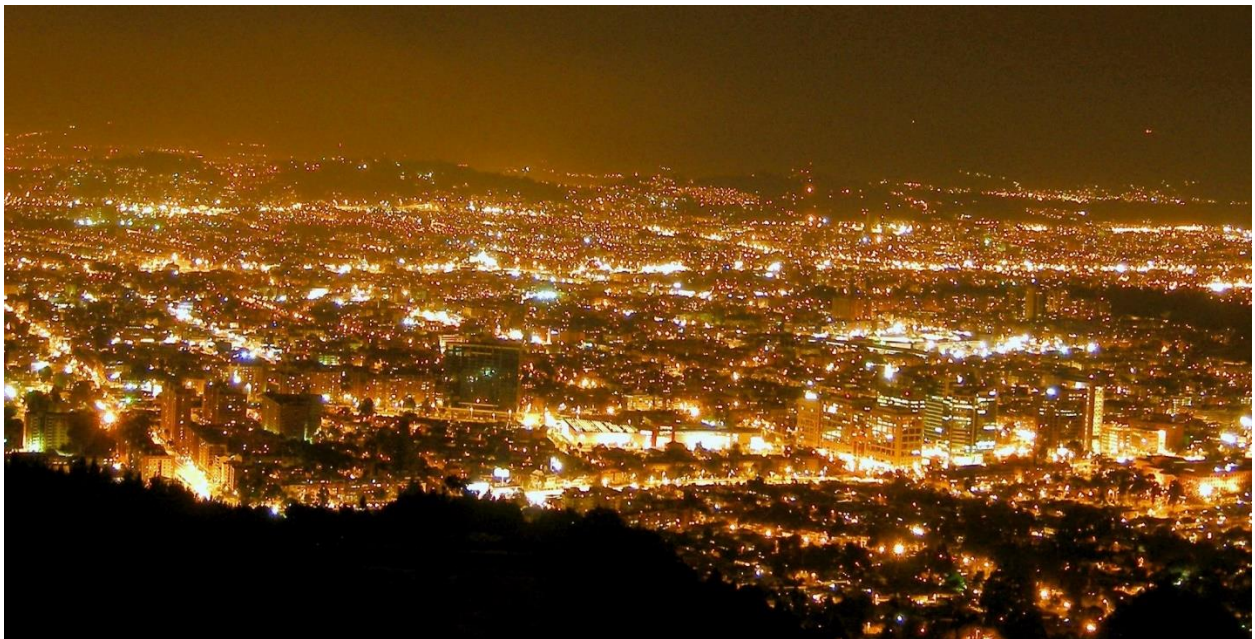
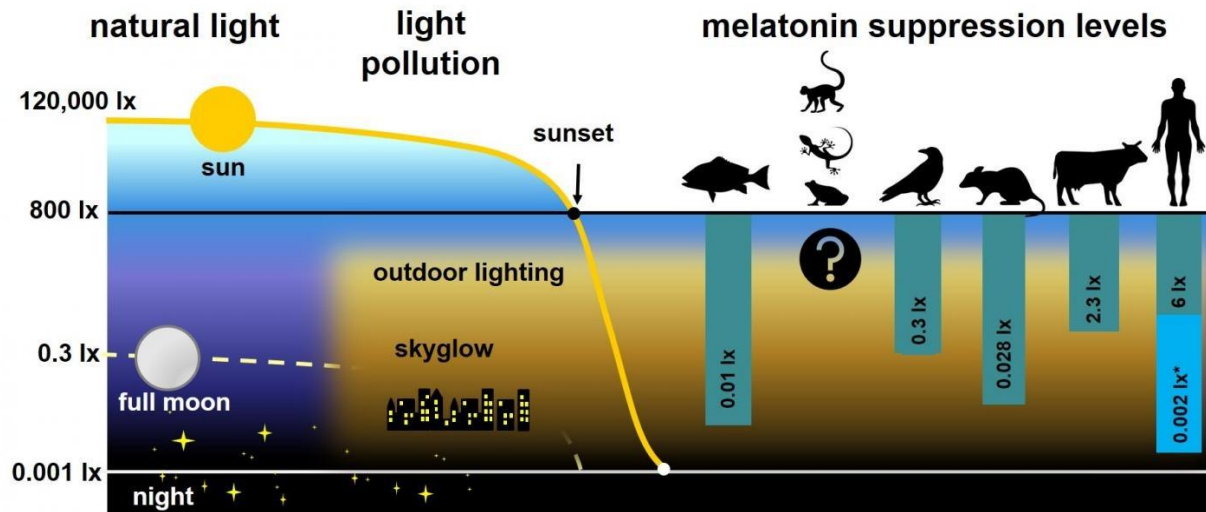


Photo credit: <https://therevelator.org/cities-ranked-light-pollution/>

The development of artificial lighting has completely changed how we work, play, and live. Artificial light has permeated every aspect of our life, from Thomas Edison's first light bulb to today's LED lights. However, the issue of light pollution has gotten worse as a result of our increased reliance on artificial lighting. Astronomer David L. Crawford originally used the term "light pollution" in the 1970s. Crawford coined the term to illustrate how artificial illumination interferes with astronomical views and the natural world.

Light pollution is generally understood to be excessive or improperly focused artificial light that interferes with the night sky's normal blackness. It is a result of the growing use of artificial light sources, which influence not only human activities but also the natural environment and animals. These sources include streetlights, outdoor security lights, and advertising signs. Light pollution may harm the ecosystem in a number of ways, including by interfering with astronomical observations, hurting nocturnal pollinators, and messing with plants' and animals' circadian cycles.

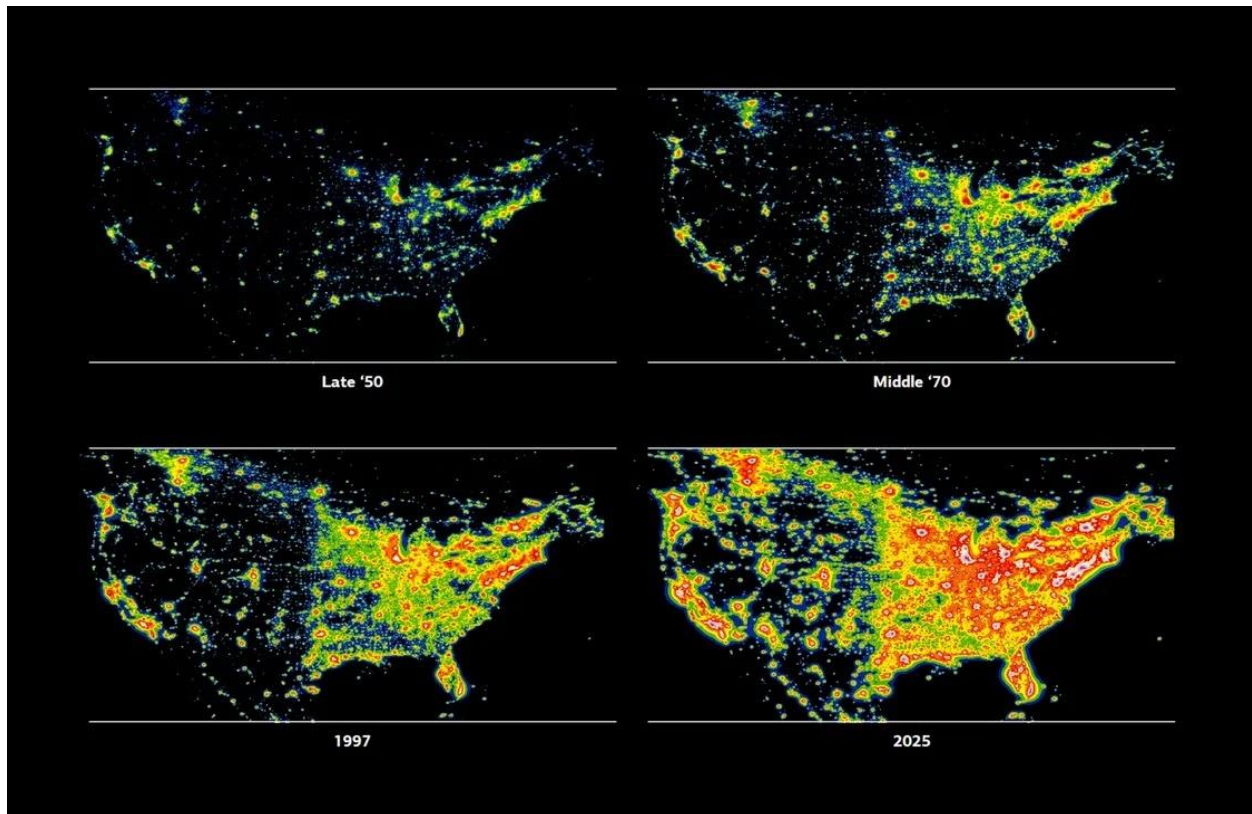
Light pollution can harm animals, especially birds, insects, and sea turtles. According to research in the UK, locations with high levels of light pollution have a 45% lower species richness than those with low levels of light pollution, which is associated with a fall in moth populations (Davies et al. 2013). Other species that eat these insects might suffer as a result of the decline in moth populations. Similarly to this, a different study discovered that light pollution interfered with migratory bird behavior, increasing mortality rates and population reduction. One of the most well-known impacts is the disorientation of sea turtle hatchlings, which are drawn to man-made lights on land rather than the moonlight ocean, resulting in high fatality rates (Longcore and Rich 2004). Light pollution has been linked to higher death rates and population reduction in migratory bird species. According to research, artificial nighttime light causes migrating birds to get disoriented and crash into houses and other objects (Loss et al. 2014).



The link between light (illuminance levels) & melatonin suppression in both humans and animals.

Photo credit: Andreas Jechow, Icons made by freepik

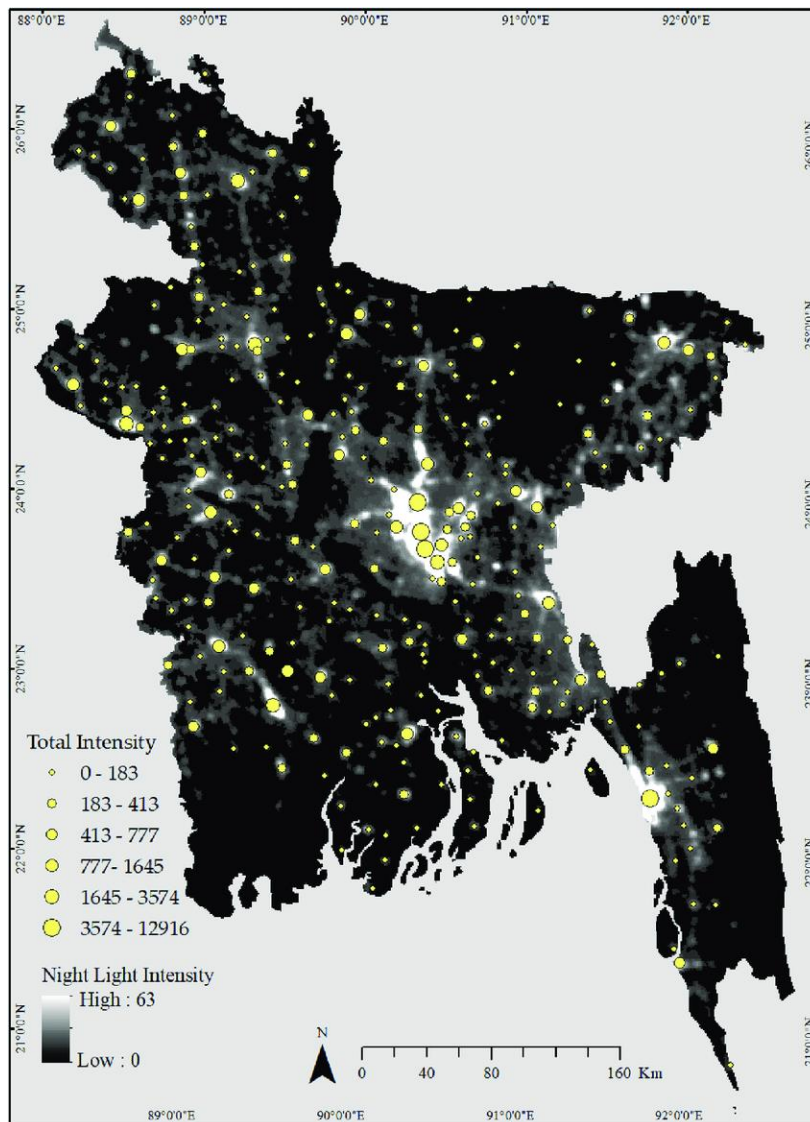
In addition to harming the environment, light pollution may also have detrimental effects on human health. The effect on our sleep habits is one of the main issues. Artificial lighting at night has the potential to interfere with our circadian rhythm, which controls our sleep-wake cycle and causes sleep problems such as insomnia, sleep deprivation, and frustration. According to research (Czeisler et al. 2015), this disturbance of sleep patterns can lead to major repercussions including impaired cognitive performance, emotional problems, and a higher risk of accidents and injuries. Additionally, studies indicate that exposure to artificial light at night may increase the risk of developing chronic illnesses like diabetes, cardiovascular disease, and cancer. The interruption of melatonin synthesis, a crucial hormone that controls our body's internal clock and has been related to the prevention of various illnesses, is likely to be the cause of this (Stevens et al., 2017). Also, macular degeneration, a condition that can result in blindness, has been linked to an increased risk of development due to the blue light emitted by LED lighting, which is frequently used in outdoor lighting (Bergen et al., 2019).



A view of the night sky is being taken away from us by contemporary living, as seen by the exponential development of light pollution.

Photo credit: <https://archive.curbed.com/2017/4/26/15442432/photography-dark-sky-light-pollution-skyglow>

The impacts of light pollution are particularly noticeable in Bangladesh. Due to the growing use of artificial lighting at night as a result of the country's urbanization, the natural cycle of light has been disrupted, and light pollution has increased. This has caused nocturnal animals to decline, including bats and owls, which are light-sensitive and depend on nighttime darkness for mating and hunting. Additionally, it has interfered with bird migration patterns and caused fish populations in wetlands and rivers to decline because of the excess light. However, the effects of light pollution do not just affect the natural world. Additionally at risk is human health. A number of health issues, including disturbed sleep patterns, an elevated risk of cancer, and mental health disorders including despair and anxiety, have been related to exposure to artificial light at night. The health effects of light pollution might have significant repercussions in Bangladesh, where access to healthcare is scarce.



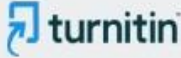
Bangladeshi cities' combined nighttime light intensity in 2001.

Photo credit: https://www.researchgate.net/figure/Composite-night-light-intensities-of-cities-in-Bangladesh-in-2001_fig1_329807469

Even while light pollution has drawbacks, there are ways to lessen its effects. Adopting lighting designs that are compatible with the dark sky reduces light pollution by directing light only where it is required and preventing unnecessary light spillage. Energy-efficient lighting solutions, such as LED lights, which consume less energy and produce less light pollution, can be used to achieve this. In addition to technological adjustments, public outreach, and education are essential in the fight against light pollution. Citizens may be involved in monitoring light pollution and advocating safe lighting practices through community-based projects like citizen science activities. Citizens

may feel a feeling of duty and ownership as a result, leading to an ongoing effort to remedy the problem. Finally, changes in law can offer a framework for regulating light pollution, such as dark-sky laws and zoning laws for outdoor lighting. These regulations can impose restrictions on light pollution, require the use of lighting systems that are friendly to the dark sky, and offer incentives for the use of energy-efficient lighting technology.

In conclusion, the effects of light pollution on the environment, animals, and public health are important, and they are becoming an increasing problem in Bangladesh. A multi-pronged strategy is needed to address this problem, including the use of lighting schemes that are friendly to the dark sky, public outreach and education, and governmental changes. We can assure a sustainable future for Bangladesh's ecosystems and people populations if we work together to address this problem.



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Learnings and Outcomes (Self-reflection):

- I learned how to write a feature article supported by research and evidence.
- I can now explain how light pollution affects both animals and human health.
- I learned how to connect global research with local Bangladeshi context.